

Efficacy and Safety of Sitagliptin, Metformin and Glimepiride FDC Tablets in Indian Type 2 Diabetes: A Subgroup Analysis of Patients Aged <50 Years

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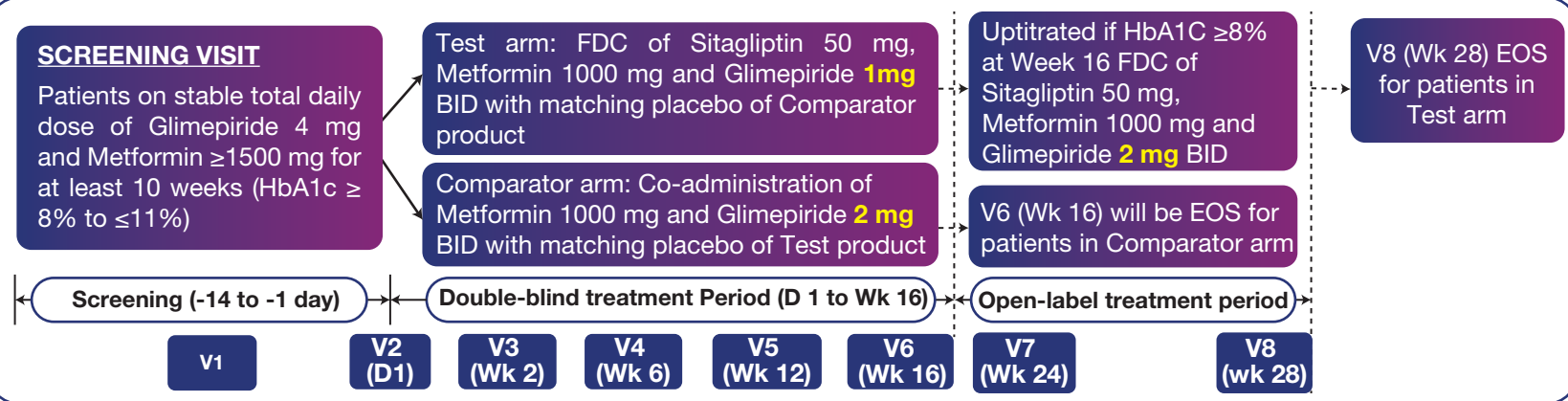
Background

- Findings from the NFHS estimates prevalence of T2DM in the age group of 35-49 years as 9.19%. This follows the trends observed in Asians and Indians that T2DM occurs at a younger age and at a lower BMI.^{1,2}
- Use of FDC of DPP4i, sulfonylurea and Metformin may improve treatment compliance and provide superior glycemic control, leading to better patient outcomes.
- We present here subgroup analysis (n=230) of patients with age <50 years at Week 16 of a phase 3 randomized, double-blind study (N=392).

Methods

- This is a subgroup analysis of Phase 3 study (CTRI/2021/11/038169) Sitagliptin+Glimepiride+Metformin FDC (50/1/1000 mg) in patients with T2DM.
- This study included patients with HbA1c ≥8% to ≤11% and received either FDC of Sitagliptin+Glimepiride+Metformin tablet (50/1/1000 mg) BID (Test) or co-administration of Metformin 1000 mg BID and Glimepiride 2 mg BID (Comparator) for 16 weeks.

Study Design



- For this analysis, individuals were categorized into subgroups (S1 and S2) based on baseline age (S1: age ≥40 to < 50 years; S2: age <40 years).

Results

Baseline Characteristics

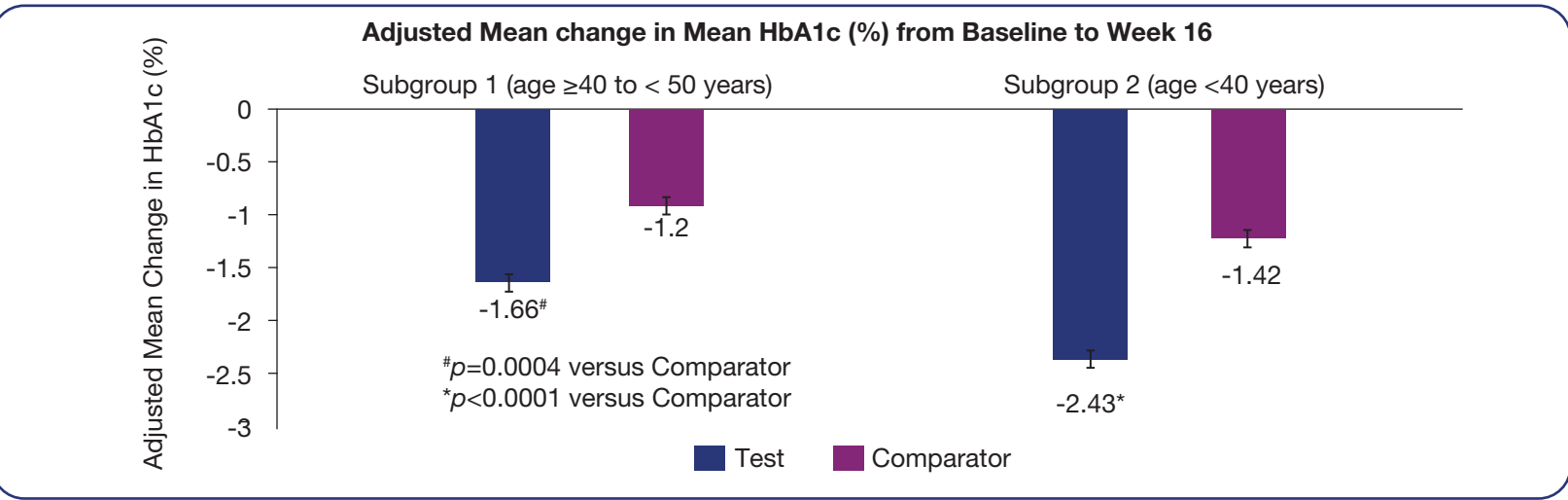
- In this subgroup analysis, a total of 150 (70 in Test and 80 in Comparator) patients were included in S1 and 80 (37 in Test and 43 in Comparator) patients were included S2.
- Baseline characteristics were comparable between Test and Comparator arms in both the subgroups.

	Subgroup 1 (age ≥40 to < 50 years)		Subgroup 2 (age <40 years)	
	Test (N=70)	Comparator (N=80)	Test (N=37)	Comparator (N=43)
Gender	n (%)			
Male	42 (60.0%)	51 (63.8%)	18 (48.6%)	27 (62.8%)
Female	28 (40.0%)	29 (36.3%)	19 (51.4%)	16 (37.2%)
	Mean±SD			
Age (years)	44.17 ± 2.86	44.26 ± 2.71	35.76 ± 3.25	34.88 ± 3.66
BMI (kg/m²)	25.41 ± 3.26	25.50 ± 3.35	25.01 ± 3.17	26.06 ± 4.49
HbA1c (%)	9.16 ± 0.73	9.10 ± 0.72	9.27 ± 0.76	9.10 ± 0.74
FBG (mg/dL)	183.4 ± 39.96	182.9 ± 38.12	176.6 ± 34.43	189.5 ± 32.97
PPBG (mg/dL)	257.3 ± 44.96	250.2 ± 54.98	237.2 ± 47.96	250.9 ± 40.38

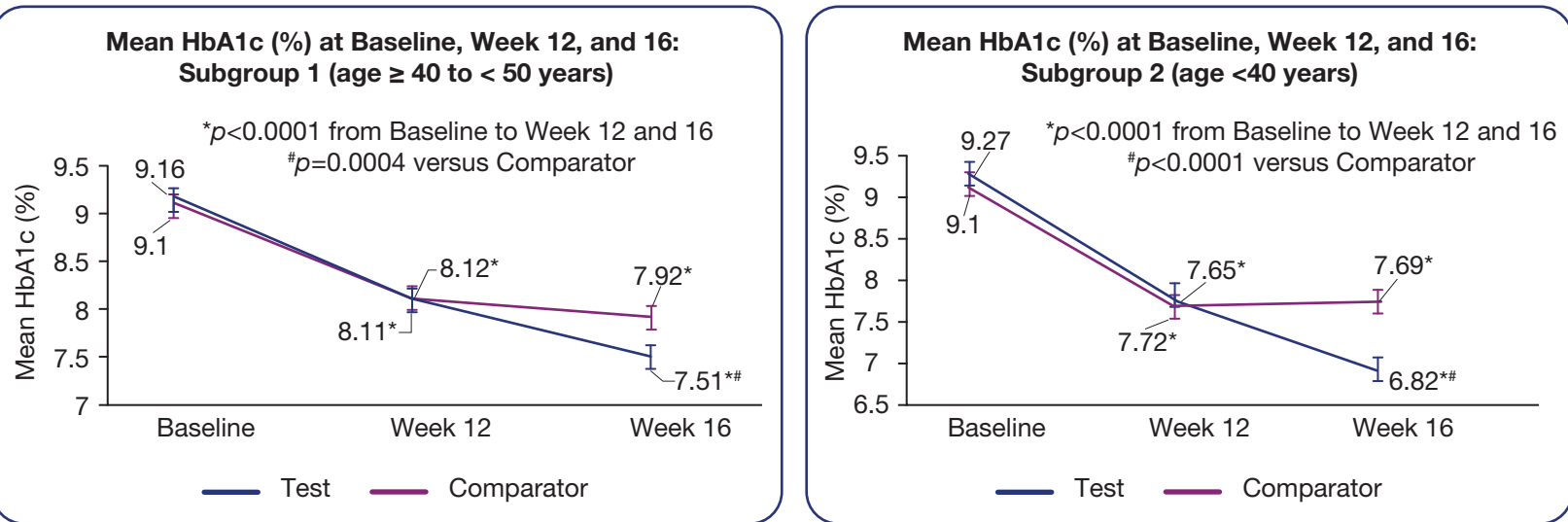
Test: FDC of Sitagliptin, Metformin and Glimepiride tablets (50 mg/1000 mg/1 mg) BID Comparator: Co-administration of Metformin 1000 mg BID and Glimepiride tablets 2 mg BID

Change in HbA1c (%) from Baseline

- In both subgroups, Test arm demonstrated statistically significant reduction in adjusted mean change in HbA1c (%) as compared to Comparator arm at Week 16 (S1: $p=0.0004$; S2: $p<0.0001$).



- Week 12 and 16, patients in both arms in both subgroups demonstrated significant improvement in HbA1c from Baseline ($p<0.0001$ each).



Change is FBG and PPBG from Baseline to Week 16

- A statistically significant reduction in FBG from Baseline to Weeks 6, 12 and 16 were observed in both arms in both subgroups ($p<0.0001$ each).
- A statistically significant reduction in FBG from Baseline to Weeks 16 was observed in Test versus Comparator arm in S1 ($p=0.0304$).

Change is FBG (mg/dL)

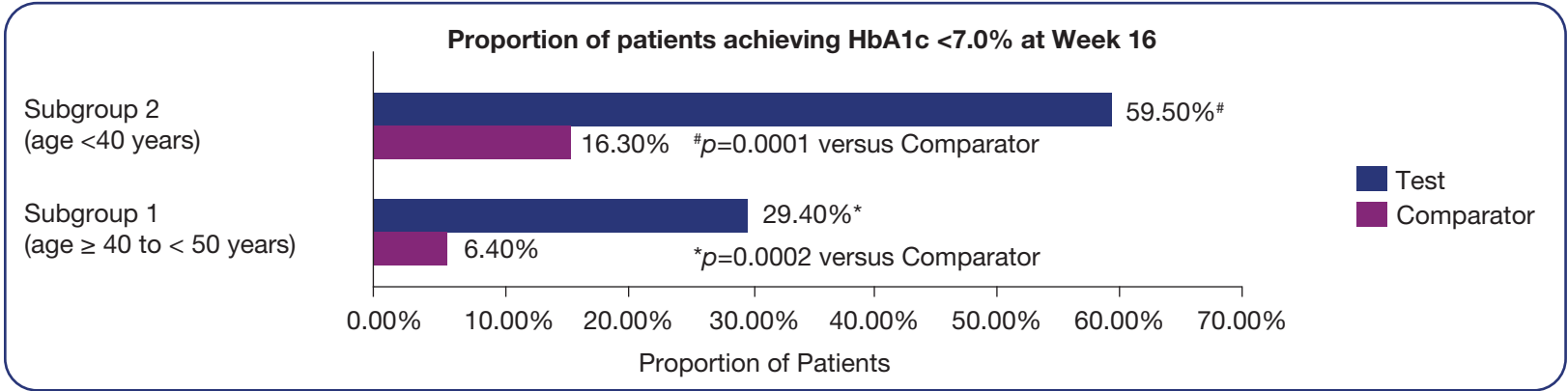
Test product	Test	Comparator	Test Vs Comparator
Subgroup 1 (age ≥40 to < 50 years) (Mean±Standard Deviation)			
FBG at Baseline	180.4 ± 38.03	179.3 ± 41.55	0.8731
FBG at Week 16	132.2 ± 34.96	142.5 ± 31.63	0.0304
Change from Baseline	-47.7 ± 41.72	-36.6 ± 41.71	
p value	<0.0001	<0.0001	
Subgroup 2 (age <40 years) (Mean±Standard Deviation)			
FBG at Baseline	174.3 ± 33.48	188.3 ± 35.64	0.0762
FBG at Week 16	132.7 ± 26.95	137.5 ± 26.83	0.6928
Change from Baseline	-41.6 ± 24.53	-50.8 ± 29.98	
p value	<0.0001	<0.0001	

Change is PPBG (mg/dL)

In both subgroups (S1 and S2), reductions in PPBG from Baseline to Week 16 were statistically significant in both Test and Comparator arms ($p<0.0001$ each) and were comparable between arms.

Proportion of Patients Achieving HbA1c <7.0%

In both S1 and S2, proportion of patients achieving HbA1c <7.0% was statistically significant in Test versus Comparator (S1: 29.4% vs 6.4%, $p=0.0001$; S2: 59.5% vs 16.3%) at Week 16.



Adverse Events – Overall Summary

- Safety profile was comparable between both arms in both subgroups, and the treatments were safe and well-tolerated.
- Total 31 TEAEs (10 in Test and 21 in Comparator arm) in S1 and total 6 mild TEAEs (3 in Test arm: unlikely related and 3 in Comparator arm: 1 unlikely related and 2 probable/likely related) in S2 were reported.
- Of the 31 TEAEs in S1, 26 were mild (10 in Test and 16 in Comparator arm), 4 were moderate (in Comparator arm) and 1 was severe (in Comparator arm).
- The most common TEAEs in S1 included pyrexia (1 in Test arm and 5 in Comparator arm), vomiting (3 in Test arm and 2 in Comparator arm), and nausea (4 in Comparator arm.)
- The TEAEs in S2 included diarrhea, pyrexia, and headache in Test arm and abdominal pain upper, nausea, and headache in Comparator arm.
- Two hypoglycemia events were reported in S1 in Test arm, which required no intervention.
- In this study, no serious AEs were reported; no discontinuations due to AEs; and all TEAEs were resolved.
- None of the patient required rescue medications during the study.

DISCUSSION AND CONCLUSION

- Irrespective of the age at Baseline, Test product demonstrated statistically superior efficacy as compared to the Comparator product.
- A statistically significant reduction in glycemic parameters were observed from Baseline to end of Week 16 in both the treatment arms.
- Proportion of patients achieving HbA1c <7.0% was statistically significant in Test versus Comparator at Week 16 in both the subgroups.
- Thus, the synergistic mechanism of action of Sitagliptin in combination with Metformin demonstrated an optimum glycemic control with a lower dose of Glimepiride in patients who were previously on a higher dose of Glimepiride.
- Safety profile was comparable between both the arms and both the study treatment were safe and well-tolerated.

Abbreviations:
AE: Adverse events; BID: Twice daily; BMI: Body mass index; D: Day; DPP4i: dipeptidyl peptidase-4 inhibitors; EOS: End of study; FBG: Fasting blood glucose; FDC: fixed-dose combination; HbA1c: Glycated hemoglobin; NFHS: National Family Health Survey; PPBG: Postprandial blood glucose; T2DM: Type 2 diabetes mellitus; TEAE: Treatment emergent adverse events; V: Visit; Wk: Week

References:
1. Chandrupatla SG, et al. *Epidemiol Health*. 2020;42:e2020065. 2. Siddiqui MK, et al. *Diabetologia*. 2022 Jun;65(6):973-983.